EXPERIMENT-1

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<u>Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on use memory.</u>

<u>FRAME 1:</u>

<u>INSTRUCTION PAGE</u>

Chunking Analysis of the Instruction Page

Chunking is a cognitive technique that organizes information into smaller, meaningful units, making it easier for users to understand and remember instructions. The Emoji Memory Game instruction page effectively applies chunking in the following ways:

1. Clear and Sequential Instructions

- The instructions are presented as short, numbered steps.
- Each step focuses on one clear action, reducing confusion and cognitive load.
- Users can easily understand the flow of the game before starting.

2. Logical Grouping of Information

- **Viewing Phase** – Users are informed that emojis will be displayed.
- **Memorization Phase** – Users are encouraged to focus on remembering emojis.
- **Recall Phase** – Users are guided to select the emojis they remember.

- **Performance Focus** – Accuracy is emphasized over speed.

3. Visual Hierarchy and Design Support

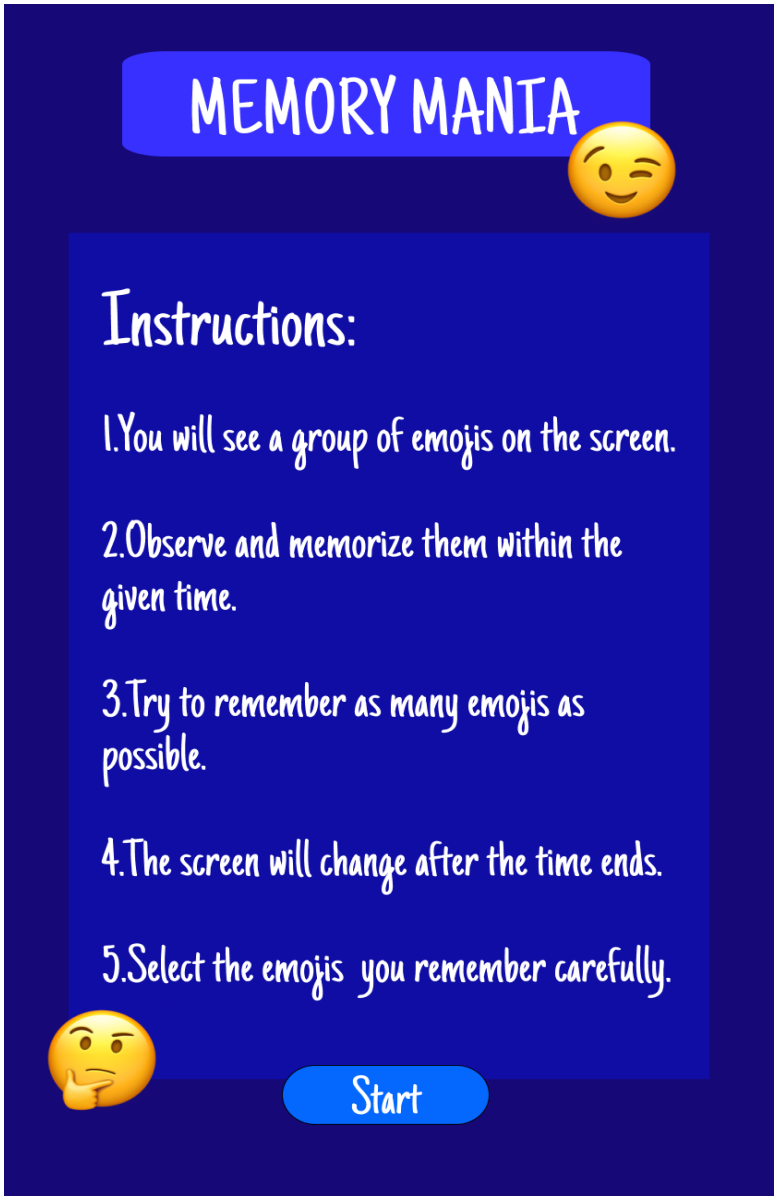
- A bold and prominent title clearly communicates the task.
- Proper spacing and alignment improve readability.
- A highlighted **START** button clearly indicates the next action.
- Emoji and mascot elements add a playful tone without overwhelming the content.

4. Time Expectation Clarity

- The instruction page clearly informs users about the limited viewing time.
- This prepares users mentally and sets clear expectations for gameplay.

5. Simplicity and Readability

- Instructions use short, simple sentences.
- Active voice keeps the content engaging and action-oriented.
- Minimal text ensures quick understanding.



Prototype Link:
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FRAME 2:

CHUNKING PHASE

Analysis of the Emoji Memory Game – Chunking Phase Screen

This screen represents the memory encoding stage, where users observe emojis and organize them mentally within a limited time.

1. Purpose of the Screen

- Allows users to visually observe and memorize emojis.
- Encourages the use of chunking by grouping emojis based on similarity.
- Supports short-term memory encoding.

2. Key UI Elements

⏱ Countdown Timer

- Displays the remaining viewing time.
- Creates urgency and keeps users focused.

□ Emoji Grid

- Emojis are arranged in a structured grid layout.
- Visual grouping helps users form memory chunks based on:
    - Emoji type (animals, food, objects).
    - Color and visual similarity.
    - Repeated emoji patterns.

🐝 **Mascot Character**

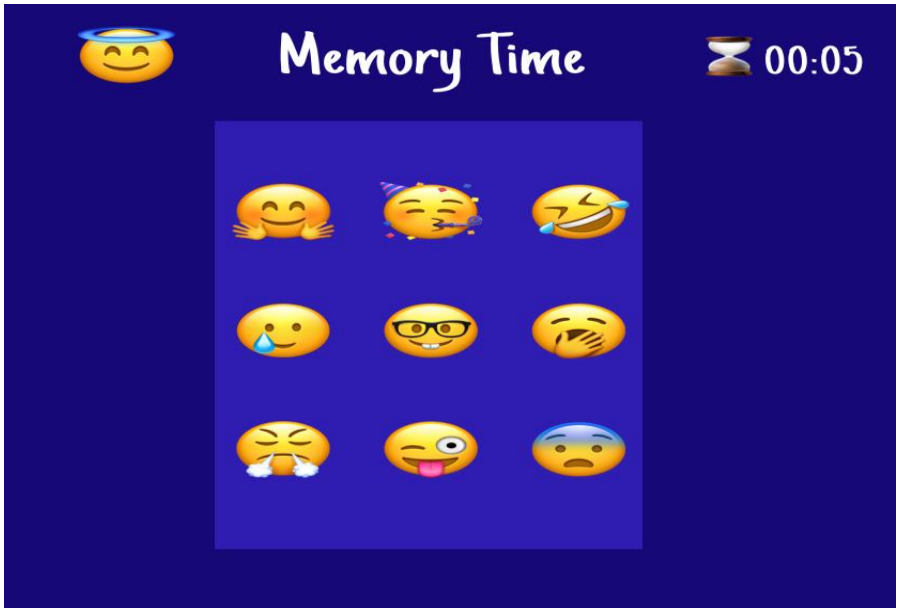
- Adds a friendly and engaging visual element.
- Helps maintain user interest during the timed phase.

3. Chunking Mechanism in Action

- Users scan the grid and mentally group related emojis.
- Limited time encourages quick pattern recognition.
- The screen transitions automatically to the recall phase.

4. Cognitive Benefits

- Enhances memory by grouping information.
- Reduces overload by organizing visual data.
- Improves recall accuracy.



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FRAME 3:

RECALL PHASE

Analysis of the Emoji Memory Game – Selection Phase

This screen tests the user's ability to retrieve stored information from short-term memory.

1. Purpose of the Screen

- Measures how effectively users remember emojis from the chunking phase.
- Tests recall accuracy using recognition-based selection.

2. Key UI Components

📝 **Title and Instructions**

- Reinforces the task goal.
- Provides clear guidance on what the user must do.

□ Emoji Options Grid

- Includes emojis seen earlier along with distractor emojis.
- Users must identify correct emojis based on memory.
- Selection controls ensure clear and deliberate choices.

🦁 **Mascot Element**

- Maintains visual continuity and engagement.

□ Submit Button

- Allows users to finalize their recall choices.

3. How Recall Works

- Users compare current emojis with stored memory.
- Distractors test memory precision.
- Submission leads to performance evaluation.

4. Cognitive and UX Benefits

- Tests true recall instead of guessing.
- Improves memory discrimination skills.
- Simple interaction keeps the process stress-free.

Prototpe Link:
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FRAME 4:

RESULT PAGE

Analysis of the Emoji Memory Game – Score & Feedback Screen

This screen provides feedback and reinforces learning through performance evaluation.

1. Purpose of the Screen

- Displays recall accuracy clearly.
- Motivates users to continue or retry the game.

2. Key UI Components

📝 Score Display

- Shows correct answers in a clear numeric format.
- Helps users understand their performance instantly.

□ Action Buttons

- **Restart** – Allows replay for improvement.
- **Exit** – Ends the game.

🐝 Mascot Character

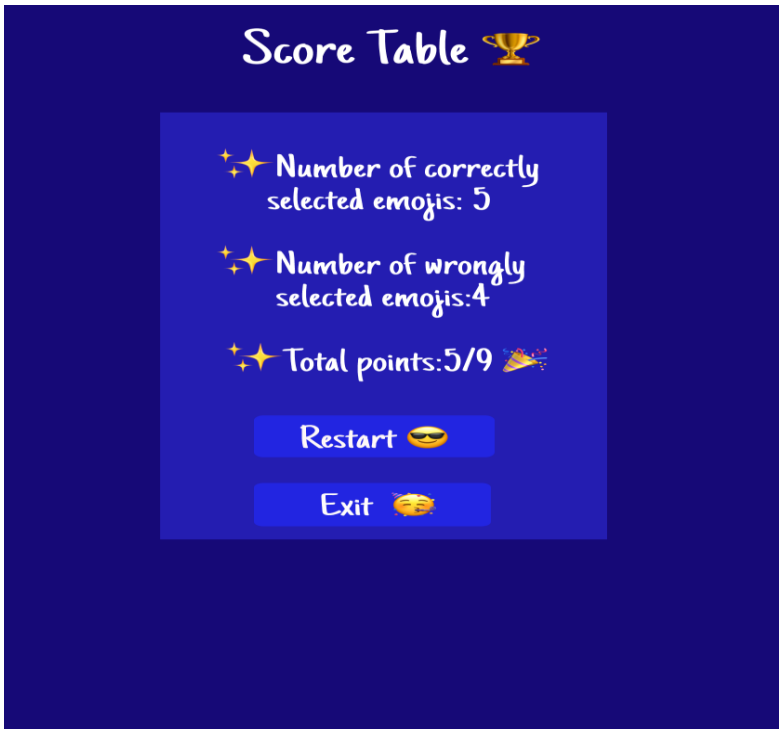
- Adds emotional positivity to feedback.
- Makes results feel encouraging rather than stressful.

3. How This Phase Works

- The system calculates correct responses.
- Results are displayed visually.
- Users choose their next action.

4. Cognitive and UX Benefits

- Immediate feedback reinforces learning.
- User control improves engagement.
- Gamified visuals reduce anxiety and increase motivation.



Prototype Link:
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